

114 學年度 第 1 學期 作業系統 Operating System 課程綱要

課程名稱： (中文) 作業系統 (英文) Operating System		開課單位：		資科工碩			
		永久課號：		CSIC30015			
		上課時間/教室：		W34-EC122[GF]			
授課教師： 吳俊峯							
學分數：		3.00	必 / 選修：		選修	開課年級：	*
先修科目或先備能力： Welcome beginners!							
課程概述與目標： This course is an advanced version of the undergraduate course "Introduction to Operating System". Generally, most topics are the same as those of the undergraduate counterpart, but we will dive deeper to explore how are those topics implemented in Linux and what is the trend or challenge for each topic. For each topic, we will revisit key concepts provided in the undergraduate OS, and then move to advanced topics, mostly from selected papers. We will also have some hands-on projects on Linux, such as playing with kernel system call and CPU scheduling.							
教科書 (請註明書名、作者、出版社、出版年等資訊)：		1. Linux Kernel Development (3rd Edition), by Robert Love 2. Professional Linux Kernel Architecture, by Wolfgang Mauerer 3. Top System Conference Venues: USENIX OSDI, USENIX NSDI, USENIX FAST, USENIX ATC, ACM/IEEE ISCA, IEEE MICRO, IEEE HPCA, ASPOLOS					
課程大綱			分配時數			備註	
單元主題	內容綱要		講授	示範	習作		
教學要點概述：							
1.學期作業、考試、評量： (tentative, subject to change without notice) midterm - 20%							

final - 30%

paper presentations -20%

hands-on projects -30%

attendance, class participation, etc. - 5%

2.教學方法及教學相關配合事項 (如助教、網站或圖書及資料庫等)

師生晤談	排定時間	地點	聯絡方式
	TBA	TBA	TBA

每週進度表

週次	上課日期	課程進度、內容、主題	教師授課時數
1	2025-09-03(三)	Course Introduction	
2	2025-09-10(三)	Operating Systems a Recap   Challenges	
3	2025-09-17(三)	Process  CPU Scheduling: Process  CPU Scheduling	
4	2025-09-24(三)	Process  CPU Scheduling: CPU Scheduling, Interrupts	

5	2025-10-01(三)	Main Memory Management: Memory Hierarchy, Virtual Memory, Paging, Buddy System	
6	2025-10-08(三)	Holiday Week, No Class	
7	2025-10-15(三)	Advanced Main Memory Management: Rethink context switch, Rethink process scheduler, OS-Controlled Memory  Hardware-Controlled Memory Extension	
8	2025-10-22(三)	Midterm Exam	
9	2025-10-29(三)	I/O Devices  Device Driver: HDD, SSD, SMR, Block I/O Layer	
10	2025-11-05(三)	I/O Devices  Device Driver: HDD, SSD, SMR, Block I/O Layer	
11	2025-11-12(三)	File System: Virtual File System	
12	2025-11-19(三)	File System: Virtual File System	
13	2025-11-26(三)	File System: Organization, Interface and Implementations	
14	2025-12-03(三)	Break, No Class	
15	2025-12-10(三)	Advanced Topic: In-Memory File System  Key-Value Databases	
16	2025-12-17(三)	Final Exam	

備註：

1.請遵守智慧財產權觀念及勿使用不法影印教科書。

2.其他欄包含參訪、專題演講等活動。

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